

Customising the ASCII report

The ASCII report can be customised based on the following:

Bill of materials. The ASCII report will be sorted based on factors such as component name, type of materials, consumables (for example, solder and solder paste) and routing information. Useful for determining the amount of each component required for the designing.

Part list. This list is for browsing only the top-level components and avoids material, routing information and consumables.

Component location. Gives you details of the position of each component—its position and angle in X-Y axes, the side of PCB on which they are placed (top or bottom).

PickPlace location (ppl). This file is required to automatically assemble the PCB according to the X-Y coordinates and relative positions.

Jumpers info (txt). Used to find out the jumpers used in the design.

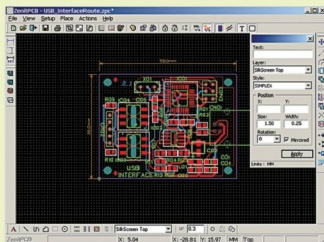


Fig. 2: A complete PCB design using ZenitPCB

(Sigma - Σ , Pi - Π , bidirectional arrows - $\leftrightarrow$, hysteresis - $\bar{\bar{J}}$ etc), off-page connector symbols, wires, shapes, texts, field texts and so on in the schematic. Other features include auto-renaming of reference, ability to mirror the parts, customisable colour schemes and visibility for each of the layers, availability of datasheets on parts provided in the library, option to export netlist format files into PCB and customisable ASCII report.

ZenitPCB allows the export of reports into ASCII files into either no separation (as image) or separated (comma-separated values or CSV - to import into spreadsheet).

ZenitPCB layout tool lets you use Netlists from competing software

This tool allows the user to import a netlist file from various other major schematic capture software (OrCAD by Cadence Design Systems, Protel/Altium

Designer by Altium Limited, EAGLE by CadSoft Computer, NI Multisim by National Instruments Electronics Workbench Group, PADS by Mentor Graphics, to name a few) while preserving all the components and their relative electrical connections.

Design without schematic. It is also possible to

create the PCB design project directly from the layout editor, without the need for importing netlist from schematic capture. This can be done by importing components from the library and then connecting those pins using the cursor (auto ratlines).

Jumpers. Jumpers are short pieces of conductors that let you bypass part of a circuit and make electric connection between two pin headers. The current version has an autojumper introduced to it. When routing, you can import jumper decals into the project and place the selected surface-mount jumper on the top or bottom side of the PCB.

Photo View. This view gives you the idea of how the real PCB would look like; it shows all the through holes, slots and silk screens. The user can check whether the components are placed in the correct side, the silk and references are placed well or end up on the pads.

Netlist format is basically an ASCII (American Standard Code for Information Interchange) text file that contains electrical and electronics information.

Signal trace is the copper foil that remains after etching a PCB. It conducts signals and is equivalent to connection wires.

Rename Components. The AutoRename Reference command allows you to rename all the components in the layout. You could choose whether they should be numbered horizontally or vertically, and the width of the band. Also, you will be able to create ASCII for back annotations of several schematic captures.

Import/Export DXF. It is possible to import and export every layer and associated items from and to DXF files compatible with commercial CAD such as AutoCAD, IntelliCAD and TurboCAD.

Export IDF. It is possible to export the design created in ZenitPCB to ASCII file containing all the 3D information. This file is compatible with most professional CAD software, making it useful for mechanical designers.

Trace Current Capacity. The Current Capacity option lets you know how many amperes of current a trace could carry, by choosing the empirical formula, foil thickness and temperature rise above ambient condition.

Alternative Bottom Component. This field in component library lets you set the bottom footprint for both wave soldering and reflow soldering.

Obstruct Shapes. It is possible to define areas in the PCB where certain operations such as copper pour, placement and routing are not allowed. Height obstruct option is used for checking against component keep-out areas.

Padstack. New features allow the user to make pad in an offset shape, and with slot holes.

You can make both single-sided and double-sided PCBs using ZenitPCB Layout. The component library has more than thousand footprints. The footprint wizard makes the creating of